

Average Trap Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	average-trap

Big idea: An average is a choice of representation, not a neutral summary of reality.

Student-Facing Prompt

Inspect Average Trap Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Average Trap Stimulus A	Student-facing stimulus 1: Revision hours: 1, 1, 2, 2, 3, 3, 18. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Average Trap Stimulus B	Student-facing stimulus 2: Screen time claim: mean looks high, median looks moderate, mode shows the most common routine. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Average Trap Reveal / Twist	Student-facing stimulus 3: Screen time claim: mean looks high, median looks moderate, mode shows the most common routine. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — An average is a choice of representation, not a neutral summary of reality.

Actual Lesson Questions

#	Question for students
1	After inspecting Average Trap Stimulus A, what is your first knowledge claim?
2	Which exact detail from Average Trap Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	How does choosing a summary statistic shape knowledge?

Ready-to-Use Examples

- Revision hours: 1, 1, 2, 2, 3, 3, 18.
- Screen time claim: mean looks high, median looks moderate, mode shows the most common routine.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Give students a skewed dataset and ask them to describe a typical value.
2. Groups calculate or identify the mean, median, and mode.
3. Students match each average to a possible headline or policy claim.
4. Groups decide which average is most responsible for the audience and purpose.
5. Debrief why one correct number can create a partial picture.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: When does a mathematical summary become misleading?
- Who decides which average is most relevant?
- Can a number be accurate but unfair?

Knowledge Questions

- How does choosing a summary statistic shape knowledge?
- What makes a mathematical representation fair?
- When is simplification necessary, and when is it distortion?

Transfer Example

For example, run this activity with a simple classroom version using small datasets for incomes, screen time, revision hours, or class scores.. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...